

Media Frenzies in Markets for Financial Information

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Emerging equity markets witness occasional surges in prices (frenzies) and cross-market price dispersion (herds), accompanied by abundant media coverage. An information market complementarity can explain these anomalies. Because information has high fixed costs, high volume makes it inexpensive. Low prices induce investors to buy information that others buy. Given two identical assets, investors learn about one; abundant information reduces its payoff risk and raises its price. Transitions between low-information/low-asset-price and high-information/high-asset-price equilibria resemble frenzies. Equity data and new panel data on news coverage support the model's predictions: Asset market movements generate news and news raises prices and price dispersion. (JEL D82, G12, G14)

The search for the source of frenzies and herds is a search for complementarities in asset demand: How can one person buying an asset make the asset more attractive to other investors? Perhaps the complementarities lie not in the demand for assets, but in the demand for information used to price the asset.

Why look to information markets as the source of frenzies and herds, rather than to asset markets themselves? The reason lies in the non-rival nature of information. Because assets are in fixed supply, an increase in demand must raise the market price, in order for the market to clear. This is not the case for information. An increase in demand for a piece of information in a competitive market causes more information to be provided at a lower price. This is the source of the complementarity and the source of

large fluctuations in information provision. Investors buy the same information that others are buying because that information is inexpensive. By reducing payoff uncertainty, abundant information raises an asset's price above other equally profitable assets, creating excess price dispersion (herds). Occasionally, there are large shifts in information demand as investors coordinate on a new asset to learn about. Such a shift in information suddenly raises an asset's price above what a model without information would predict (a frenzy).

The innovation in this model comes from an insight of growth theory. Paul M. Romer (1990) points out that information is fundamentally distinct from other goods because information has a fixed cost of discovery and a near-zero cost of replication. This information production technology, coupled with free entry in the information market, results in information prices that decline as demand rises. This is consistent with observed pricing schemes. Consider the relative prices of one issue of the *New York Times*, the *Economist*, *Econometrica*, and a consultant's report on one firm. The *New York Times*, the publication of most general interest, is the least expensive. As the audience becomes more specialized, prices rise.

The basis for the model is a Grossman-Stiglitz economy (Sanford J. Grossman and Joseph P. Stiglitz, 1980), where agents decide whether to purchase a signal about an asset's payoff, and then choose their asset demand, given an equilibrium price. In contrast to Grossman and Stiglitz, this model features a compet-

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itive information production sector that supplies information at an endogenous price.

Information affects asset prices and asset markets generate demand for information. Agents who purchase information observe a component of the risky asset payoff. Information reduces the asset's *conditional variance*. This conditional variance captures uncertainty—an expected distance between beliefs and the true state. In the equilibrium where information is cheap and demand is high, decreased conditional variance raises the asset's price. In the other equilibrium where information is expensive and in low demand, the asset is riskier and its price is lower. Hence, information moves asset prices. Switches between information equilibria occur because of changes in the asset payoff's *unconditional variance*. This unconditional variance measures payoff volatility. The model features a state process that ties unconditional variance to expected asset payoffs. When unconditional variance is high, information about the payoff is valuable; a positive-information equilibrium arises. With low volatility, a no-information equilibrium prevails. A media frenzy is triggered when an asset payoff increase raises unconditional payoff variance. The added unconditional variance prompts a switch to the positive-information equilibrium. Because it reduces conditional variance, abundant information raises the asset's price above what the improvement in fundamentals would predict.

In a multiple markets setting, information complementarity can increase the dispersion of prices across markets. Because high-demand information is cheap, investors want to buy information all in the same markets. Because they prefer assets they are informed about, risk-averse investors demand all the same assets. This information-based herding creates a situation where high-value markets will generate news and the news will generate a price premium for those markets. The result is high price dispersion in times when news is abundant.

The predictions of the theory are supported by empirical evidence from emerging markets. A new panel dataset of *Financial Times* stories measures information, and data from S&P 500 measure equity prices. High news levels coincide with high payoff variance and high price levels. The data also confirm the herding pre-

diction that asset price dispersion across markets is high when news is abundant.

I. A Model of Asset Demand and Information Provision

The model is in the spirit of Grossman and Stiglitz (1980). It replaces Grossman and Stiglitz's constant information price with an endogenous price set in a market characterized by increasing returns. In contrast to Anat R. Admati and Paul Pfleiderer (1986), this model features competitive information suppliers who are crucial to generating the complementarity. This model also adds a dynamic state process for the asset payoff. The state process is an important component of the model because it will cause information to be supplied in times when the asset payoff is high.

A. Preferences and Technology

Time is discrete and continues forever. In each period t , a large finite number of two-period-lived, ex ante identical agents are born. They have constant absolute risk aversion preferences over wealth (W_{t+1}):

$$(1) \quad U(W_{t+1}) = E_t[-e^{-aW_{t+1}}].$$

There are two assets: one risky asset with payoff u_{t+1} , and one riskless asset with payoff $r > 1$. The risky asset payoff has a persistent component θ and an idiosyncratic component ε :

$$u_{t+1} = \theta_{t+1} + \varepsilon_{t+1}.$$

The persistent component of payoffs is an AR(1) process with mean μ and proportional shocks η :

$$\theta_{t+1} = (1 - \rho)\mu + \rho\theta_t(1 + \eta_{t+1}).$$

Innovations are i.i.d. $\varepsilon \sim N(0, \sigma_\varepsilon^2)$, $\eta \sim N(0, \sigma_\eta)$.

A multiplicative shock to payoffs is a natural assumption in this setting. The proportionality of the mean and standard deviation is a feature of payoff processes shared by geometric Brownian motions, commonly used in asset

pricing (see Darrell Duffie, 1996). Furthermore, the assumption that payoff volatility is increasing in its level is supported in the data.¹ Realize that this assumption does not make volatility procyclical. Volatility ($\text{Var}_t(\Delta\theta_{t+1}/\theta_t)$) is constant. The only claim here is that higher-value assets are likely to see larger absolute changes in their value. That shocks are multiplicative is important because information demand will depend on the variance of payoffs. This assumption makes the variance of expected payoff innovations ($\sigma_{\theta t}$) change over time and will cause information demand to fluctuate.

B. Information

In each period t , agent i chooses whether to buy information or not. The information reveals the persistent component of next period's asset payoff (θ_{t+1}) at time t . The asset payoff u_{t+1} and the state θ_{t+1} are revealed publicly at the beginning of period $t + 1$. For any variable z , let z^t be the t -history $\{z_0, \dots, z_t\}$. Agents who purchase information at time t have an information set $\mathcal{F}_t^I = \{\theta^{t+1}, u^t, \varepsilon^t, P^t\}$ and agents who do not purchase information have access to information $\mathcal{F}_t^U = \{\theta^t, u^t, \varepsilon^t, P^t\}$.

C. Asset Markets

The risky asset has price P_t ; the riskless asset price is normalized to one. The per capita supply of the risky asset x_t is normally distributed: $x_t \sim N(E[x], \sigma_x^2)$. This supply is never observed. Agents who know θ_{t+1} , however, can infer the value of x_t from the price. The role of the supply shocks is to prevent price from perfectly revealing θ_{t+1} to uninformed agents.

D. Information Markets

The mechanism that generates herds and frenzies is price complementarity in information demand; information prices decline as de-

mand rises. Because information is nonrival, this pricing relationship arises naturally from a number of competitive market structures. This section describes one such market.

Information markets have three important features. First, information is produced according to a fixed-cost technology; θ_{t+1} can be discovered at the beginning of period t at a per capita fixed cost χ . This can be interpreted as the cost of hiring a journalist to interview people and find primary sources of information. The information, once discovered, can be distributed to other traders at zero marginal cost. Second, reselling purchased information is forbidden. The realistic counterpart to this assumption is intellectual property law that prohibits copying a publication and redistributing it for profit. Third, there is free entry. Any agent can discover information at any time.

That information markets are competitive is crucial. The exact market structure is not. Appendix A analyzes a Cournot and a monopolistic competition market as well. All three markets produce information prices that decrease in demand, complementarity in information acquisition, and frenzies.

E. Price Competition in a Contestable Market

Profits from information discovery depend on the price charged and demand for information, given the pricing strategies of other agents. The market is perfectly contestable. One way to ensure that the market is contestable is to force agents to choose prices in a first stage and choose entry in a second stage. This is a reasonable way to think of news markets where the price of the periodical is fixed well in advance and then editors decide whether or not to supply a story. By supplying the story, they would be entering the market for that piece of information.

Let $d_{it} = 1$ if agent i decides to discover information in period t and $d_{it} = 0$ otherwise. Let per capita demand for information with price c_{it} , given all other posted prices c_{-it} , be $I(\cdot, \cdot)$. Then the objective of the information producer is to maximize profit:

$$(2) \quad \max_{d_{it}, c_{it}} d_{it}(c_{it}I(c_{it}, c_{-it}) - \chi).$$

¹ In the emerging market equity data described in Section VI, a regressing time- t dividend variance (measured by squared changes) on the $(t - 1)$ dividend level yields a positive coefficient, significant at the 99-percent confidence level (t -stat = 4.05). Doing the same exercise with payoffs (next-period price plus dividends) yields a positive coefficient with a t -statistic of 16.9.

F. Order of Events

1. All agents enter the first period of life knowing the persistent component of the period t asset payoff θ_t and σ_{θ_t} , the variance of θ_{t+1} , conditional on θ_t .
 - Agents decide whether to discover information, and what price to set.
 - Agents choose whether to purchase information. Informed agents observe θ_{t+1} .
 - Informed and uninformed agents demand D^I and D^U units of the risky asset.
2. Payoffs are received. The asset supply x_t and the asset payoff u_{t+1} , with its persistent and transitory components θ_{t+1} , and ε_{t+1} are revealed.
3. In the second period of life, agents derive utility from terminal wealth W_{t+1} .

G. Equilibrium

Given an initial wealth, a persistent asset payoff process, payoff shocks, and risky asset supply, $\{\theta_t, \varepsilon_t, x_t\}_{t=1}^{\infty}$, an equilibrium is a sequence of risky asset demands and a fraction of investors who purchase information $\{D_{it}, \lambda_t\}_{t=1}^{\infty}$, and asset prices, information prices $\{P_t, c_t\}_{t=1}^{\infty}$, such that the following three conditions are satisfied in every period $t > 0$.

1. Given prices $\{P_t, c_t\}$, agents choose whether to buy information ($I_{it} = 1$), or not ($I_{it} = 0$), and choose asset demands to maximize expected utility (1), subject to their budget constraint,²

$$W_{t+1} = (W_t - P_t D_{it} - I_{it} c_{jt})r + D_{it} u_{t+1} + \pi_{it}.$$

λ_t is the fraction of agents who buy information.

2. Information supply and pricing strategies are a subgame perfect Nash equilibrium. They solve (2).
3. The markets for risky assets and information clear.

² Because of constant absolute risk aversion preferences, end-of-period information revelation, and the Markov state process, maximizing next-period utility would also be optimal for an infinitely lived agent.

II. Complementarity, Equilibrium Transitions, and Asset Prices

Most of the time, the model will produce asset prices that look just like the price in a world without information markets. Occasionally, in periods when the asset is already highly valued, information markets will supply an abundance of information, and prices will rise. Periods when information is causing the asset price to be higher than what a standard model without information would predict I will call frenzies. The mechanism responsible for the frenzies is increases in information supply that decrease the risk of purchasing the asset. Information complementarity generates large changes in the information supply. Propositions 2 and 3 show that information provision can jump between no-information and high-information equilibria. Because risk-averse investors prefer to purchase assets they are informed about, these increases in news increase the asset price, on average (Proposition 4). Positive correlation of news and asset payoffs (Proposition 3) makes frenzies happen at the right times, when asset values are already high. This positive correlation arises because information has increasing returns in the asset value.

PROPOSITION 1 (Information Price Declines in Quantity): *The equilibrium price for information $c(\lambda)$ is a decreasing function of the quantity of information purchased, λ .*

All proofs are in the Appendix. This result follows from the free entry assumption. As information demand rises, producers must produce more and reduce prices to deter new entrants. The result is more surplus for information purchasers. Price competition yields average cost pricing. Because of the fixed cost in information production, average cost declines as the number of information purchasers rises. But, the result does not depend on price competition. The Appendix shows that Cournot or monopolistically competitive prices also fall when demand rises.

To see the effect that endogenizing the information price has on information provision, examine the net benefit of purchasing news. Let the net benefit to the purchaser of the λN th signal be denoted by $B(\lambda; \theta_j)$. Grossman and

Stiglitz show that the expected net benefit of information is

$$(3) \quad B(\lambda) = \left[\frac{\text{Var}(u_{t+1}|P_t)}{\text{Var}(u_{t+1}|\theta_{t+1})} \right]^{1/2} - e^{ac(\lambda_t)}.$$

The first term is the benefit of information (plus one). When the variance of payoffs given the price ($\text{Var}[u_{t+1}|P_t]$) is large relative to the payoff variance given information ($\text{Var}[u_{t+1}|\theta_{t+1}]$), then information is valuable. The second term is the utility cost of information (plus one). The difference between the information purchasing decision in this model and in Grossman and Stiglitz is the endogenous price of information $c(\lambda_t)$, rather than a constant information price c .

PROPOSITION 2 (Information Complementarity): *At every θ_t , the expected net benefit of purchasing information $B(\lambda; \theta_t)$ is increasing in the fraction of agents that are informed λ , over some range $\lambda \in [0, \lambda_t^*]$.*

An information purchase by one agent has two opposing effects on other agents' information demand. An increase in information demand makes information cheaper and therefore more desirable to uninformed agents. This price complementarity makes the net benefit of information ($B(\cdot)$) rise in the number of informed investors (λ). Increased information provision also makes the asset price more informative ($\text{Var}[u_{t+1}|P_t]$ low), however, decreasing the value of information. The fall in the net benefit of information as the number of informed agents rises is the strategic substitutability identified by Grossman and Stiglitz (1980). In Grossman and Stiglitz, the net benefit curve is monotonically decreasing and any shifts in the net benefit curve cause information provision to change continuously. When the information price is endogenous, the net benefit curve has an increasing part. The nonmonotonic net benefit function is responsible for the large, discrete changes in information provision that inject extra volatility into the price process.

In equilibrium, information demand λ^e is such that all purchasers of information weakly prefer to have information, and no additional agents want to purchase: $B(\lambda^e; \theta_t) = 0$ (points

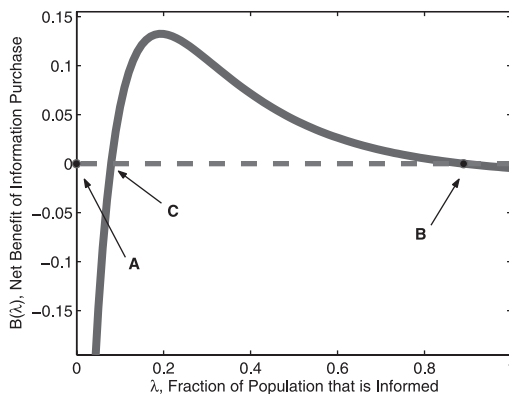


FIGURE 1. EXPECTED NET BENEFIT OF INFORMATION, AS THE AGGREGATE DEMAND FOR INFORMATION VARIES

Notes: When information supply is low, the benefit of information is increasing because its price is falling. When information is abundant, the benefit to information is declining because the asset price reveals more of the information. The parameter values used to generate the graph are: $\sigma_\theta = 0.1$, $\sigma_\varepsilon = 0.1$, $\sigma_x^2 = 10$, $a = 0.7$, $\chi = 1.8$, $n = 50$.

A, B, and C in Figure 1). If no one purchases information, the first person to buy must incur the entire fixed cost of information production. A high information fixed cost ensures that no information is an equilibrium. With a large number of agents, even a near-zero per capita fixed cost causes the cost of buying the first signal to outweigh its benefit. When many people purchase information, the cost of discovery is shared, information is cheap, and signals are a more attractive purchase.

Although A, B, and C could all be equilibria for information demand, only B is also an equilibrium in information supply. The reason is that the market in Figure 1 can support a large information supply. An agent in a state where no other agents are producing information (A) or where production is very limited (C) can make a profit by discovering information and selling it.

PROPOSITION 3 (Information Is Provided when Asset Values Are High): *For a given set of positive, finite-valued parameters, σ_η , σ_ε , σ_x , a , χ , $\{\psi_t\}$, there exists a cutoff θ^* , such that for all $\theta_t \geq \theta^*$, $\lambda_t^e > 0$ and for all $\theta_t \leq \theta^*$, $\lambda_t^e = 0$, in equilibrium.*

COROLLARY 1 (Payoff Variance Increases Information): *The fraction of informed investors λ_t is a nondecreasing function of the variance of persistent payoff innovations $\sigma_{\theta_t}^2$.*

Technically, what causes the net benefit of information for a given demand λ to fluctuate is changes in the unconditional variance of the asset payoff. When variance is high, information that reveals the level of θ_{t+1} is more valuable. Recall that the variance $\text{Var}_t[\theta_{t+1}] = \rho^2 \theta_t^2 \sigma_\eta^2$ is increasing in the level of asset payoff θ_t .³ Therefore, increases in θ_t shift the net benefit curve up. When θ_t is very low, the entire net benefit curve may lie below zero. These are times when no positive information equilibrium exists. The cutoff payoff level θ^* makes the peak of the net benefit curve lie at zero.

Intuitively, the reason that higher expected payoffs increase information demand is that information has increasing returns in the value of an asset. Because an extra piece of information is not needed to evaluate each increment of asset value, investors get more use out of information about high-valued assets. A piece of information that tells you if you will gain or lose 50 percent on an investment that is worth \$10 is not as valuable as information on whether you will gain or lose 50 percent on an investment worth \$100. All else equal, investors want to know more about higher-value markets because they have more value at risk in those markets. This relationship is not about high-value markets having more risk—percentage changes in price do not grow with asset values. It is about a given amount of risk having a bigger effect on information demand when the value of the asset at risk is large.

The dependence of information on payoff variance has two components: risk and information's increasing returns. Risk, measured as the expected squared percentage change in payoffs, increases information demand because information that predicts payoffs is most valuable when large payoff changes are likely. The mean of payoffs matters because the benefit of information increases in the value of the asset. The total variance is the risk of the asset interacted with the value of the asset at risk.

³ Large, negative asset payoffs could also generate information demand in equilibrium. This is a rare occurrence, however, because the mean of θ is above zero.

PROPOSITION 4 (Information Increases Asset Price): *The expected price of the risky asset $E[P_t|\theta_t, \lambda_t]$ is strictly increasing in the amount of information provision λ_t .*

COROLLARY 2 (Frenzies Arise when Asset Values Are High): *In periods when $\theta_t \geq \theta^*$, the expected price of the risky asset $E[P_t|\theta_t, \lambda_t]$ will be higher than the price in a world without information markets $[P_t|\theta_t, \lambda = 0]$.*

Changes in information equilibria move the asset price by changing the variance of the asset's payoffs. An expected utility maximizing agent i with information $\mathcal{F}_t^i$ who observes equilibrium asset price P_t demands $D(\mathcal{F}_t^i, P_t)$ units of the risky asset:

$$D(\mathcal{F}_t^i, P_t) = \frac{E[u_{t+1}|\mathcal{F}_t^i, P_t] - rP_t}{a \text{Var}[u_{t+1}|\mathcal{F}_t^i, P_t]}.$$

The unique market clearing price $P_\lambda(\theta_{t+1}, x_t)$ for a given information demand λ_t equates total demand for the risky asset with supply:

$$\lambda_t D(\mathcal{F}^I, P_t) + (1 - \lambda_t) D(\mathcal{F}^U, P_t) = x_t.$$

When more information is provided to the market, the conditional risk of the asset payoff, $\text{Var}[u_{t+1}|\mathcal{F}_t^I, P_t]$, is lower. Lower risk makes the asset more attractive to investors, increases demand, and raises the price. If the information predicts low asset payoffs (if $E[u_{t+1}|\mathcal{F}_t^I, P_t]$ is low), it is possible for realized prices to be lower with information than without. But on average, information is neither good nor bad. The only systematic effect of information is to reduce conditional risk and therefore increase an asset's price.

III. A Numerical Example

Any model that supplies a positive amount of information can generate a frenzy. What makes media frenzies interesting is that they have a number of realistic features: they appear only in times when the market is already highly valued, they are persistent but not permanent, they arrive and disappear abruptly, and they are large. Frenzies' appearance in booms is a result of increasing returns to information in the asset

value. Their persistence comes from linking the frenzy to the asset payoff process, rather than amplifying random shocks. Large and sudden frenzies are a result of the information market. If the information cost were instead fixed, information would be a strategic substitute; it would be more beneficial as it became more scarce. Substitutability would result in gradual variation in information demand and make it unlikely that a frenzy ever disappeared. What the complementarity in information markets does is to make the change in information provision large and abrupt. The resulting asset price path has a long series of no-frenzy prices, followed by a sudden surge in the price level, a frenzy.

One way to gauge how large the resulting frenzies are is to measure price volatility. Two features of the model inflate asset price volatility. First, switches between positive-information and no-information equilibria cause surges in the price level: large decreases in payoff uncertainty increase the price. Second, positive information equilibria exist and raise prices when prices are already high. By making the most extreme price realizations more extreme, information effects can have large consequences for the variance of prices.

Model simulations illustrate the qualitative features of media frenzies and demonstrate that the model is capable of generating frenzies of the magnitude that we see in the data.

A. Calibration

The per capita cost of information discovery χ is such that no-information and positive-information equilibria exist. The mean and variance of risky asset supply, $E[x]$ and σ_x , determine the price level and the amount of idiosyncratic noise in the price. They are chosen to be large enough to create nontrivial uncertainty about fundamentals, but not to drown out other sources of price fluctuation. Autocorrelation (ρ) of θ is equal to the weekly autocorrelation in emerging market price indices (see Section VI for data details). Variance of payoff shocks σ_ε determines the risk of investment for informed traders and the scale of the asset price. It is as large as the asset supply shocks. Absolute risk aversion a is 2.5. Finally, the variance of innovations to fundamentals σ_η is chosen to match the variance of percentage deviations of prices

from their country mean in the data and the model.⁴

B. Results: Frenzies

One measure of the magnitude of frenzies is the ratio of price variance from simulations of the model with and without information markets: $\text{Var}(P_{\text{info}})/\text{Var}(P_{\text{no info}}) = 1.5$. Information markets increase the price volatility by 50 percent. This is consistent with the finding of N. Gregory Mankiw et al. (1991) that equity prices from 1872 to 1987 exhibit excess volatility of about 50 percent. To isolate the quantitative contribution of information price complementarity, we can compare the media frenzies model to a model where information price is fixed (to the average media frenzies information price). The fixed information price model is similar to Grossman-Stiglitz but uses the dynamic state process from this model to make information abundant when asset prices are high. Simulations (10,000 iterations) show that an endogenously varying information price causes the asset price volatility to be 40 percent higher than in a fixed-information-price setting: $\text{Var}(P_{\text{info}})/\text{Var}(P_{G-S}) = 1.4$.

Figure 2 plots the price series generated by the model and a version of the model where no information is ever provided. When the information model is in a no-information equilibrium, the asset price is the same as if information markets didn't exist; the two prices lie on top of each other. It is only when there is positive demand for information that the model's prices deviate from the no-information benchmark.

The model's prices display recurrent overreactions to changes in fundamentals. For example, a 20-percent deviation of the no-information price from trend can cause a 60-percent movement in the price with information (see periods 75–80 in Figure 2). The reason is that the model breaks the link between fundamentals and prices, precisely when the changes in fundamentals are large. These large changes are most likely to move information demand. It is the shift in information demand that amplifies the price change.

⁴ The parameter values are: $\chi = 0.012$, $E[x] = 6$, $\sigma_x^2 = 1.6$, $\sigma_\eta = 0.09$, $\sigma_\varepsilon = 0.1$, $\rho = 0.99$, $a = 2.5$. The average relative risk aversion is 6.

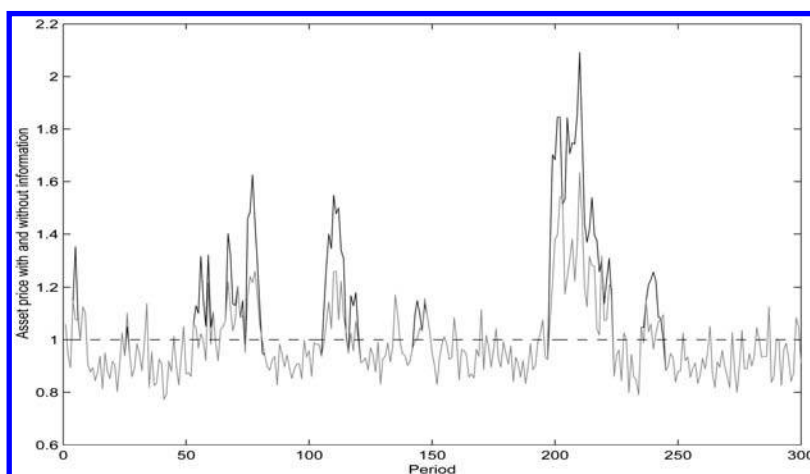


FIGURE 2. SIMULATED ASSET PRICES IN THE NO-INFORMATION MODEL (LOWER LINE) AND INFORMATION MODEL (UPPER LINE)

Note: Excess price volatility: $\text{Var}(P_{\text{info}})/\text{Var}(P_{\text{no info}}) = 1.5$.

TABLE 1—PRICE VOLATILITY IN A BENCHMARK NO-INFORMATION MODEL, IN THIS MODEL, AND IN EMERGING MARKETS

	No-information model	Information model	Emerging market data
Price standard deviation	0.29	0.39	0.43
$\text{Var}(\text{Price})$	1.0	1.8	1.3
$\text{Var}(\text{Price} \text{Low News})$			

Notes: *Low news* is defined as a level of λ , or count of news stories that falls below its median. Simulated price changes are log changes: 5,000 simulations. Emerging market data is log price indices for 23 emerging markets described in Section VI. Moments computed on a country-by-country basis, starting at country start date (see Table 2) and ending in June 2002, and then averaged across countries.

No-information prices are not directly observed, but they are functions of realized asset payoffs. Instead of constructing a sophisticated, realistic pricing model for emerging market assets, choosing a good proxy is likely to yield more reliable and less controversial results. One simple proxy is the price in periods when news provision is low. In periods when $\lambda = 0$, the prices in the models with and without information markets are the same.

Table 1 displays the unconditional price variance divided by the variance of price, conditional on observing an information level below its median. The low news price variance in the model is price variance in the 50 percent of periods when λ is lowest (λ is always zero in these periods). In the data, the low news prices are observations where the number of news

stories is less than the median of all news observations. An alternative procedure using the country median to distinguish low-news and high-news times produces a variance ratio of 1.2. The comparison suggests only that the model's effect can be large enough to be a potential explanation for frenzies.

An issue that arises in examining the asset price series is the validity of comparing this sequence of two-period asset prices to the price of multiperiod assets in the data. The difference between the assets' payoffs is price risk. While the two-period asset payoff is exogenous, the multiperiod asset payoff depends on the future expected asset price. This payoff differential is especially large near θ^* , where an information regime shift, and thus a large change in prices, is likely. Replacing the two-period asset with a multiperiod asset should

cause the price path around the cutoff point where the information regime shift occurs to be smoother. Agents in a no-information state who anticipate a likely media frenzy tomorrow will value the asset more highly because large capital gains are likely. Agents in the high-information state who anticipate the imminent end of the media frenzy will reduce their value of the asset to hedge the risk of a price fall. When the asset payoff is either a high or low outlier, however, the chance of a regime shift is small and price risk is low. In these states, the two-period and multiperiod assets will produce similar prices.

IV. Herding in a Model with Multiple Markets

In a setting with multiple investment markets, complementarity in information acquisition causes media herds. An abundance of media attention is focused on one or a subset of markets. Information on the herd market is cheap. No information is available on the other markets because any investor who wanted to purchase a signal would have to pay the entire fixed cost. The markets with information have a higher asset price because the conditional variance of their payoff is lower. When expected payoffs in a high information market start to fall, other markets may become more attractive media markets because their risk and payoff are higher. There will be a quick switch in equilibria as demand for information shifts from one market to the other. The shift in information provision causes a fall in one asset price and a rise in the other. The result is that identical markets have different equilibrium prices, and that prices exhibit excess cross-sectional volatility, relative to fundamentals.

The key mechanism to generate herds is the complementarity that arises from the information price. Some additional device is needed, however, for markets to interact with each other. Since CARA utility generates no portfolio effects between assets, each market would otherwise have an independent price process, as described by the one-market model. One simple device is to introduce a trade-off between learning about one market or another. The trade-off takes the form of a constraint on the number of signals each agent can purchase. Such a constraint could

be interpreted as limited space in newspapers or limited time to read each piece of information. Other devices that would serve the same purpose would be a budget constraint, a constraint on information processing abilities (as in Christopher A. Sims, 2003), or correlation in information so that one country's news would be an imperfect substitute for another's.

Preferences, technology, and information structure are identical to those in Section I, except for the following two modifications. First, there are M markets, indexed by m . Each market has an independent payoff process:

$$u_{t+1}^m = \theta_{t+1}^m + \varepsilon_{t+1}^m$$

$$\theta_{t+1}^m = (1 - \rho)\mu + \rho\theta_t^m(1 + \eta_{t+1}^m).$$

Innovations are i.i.d. normal, across time and markets. For simplicity, assume the shocks have common variances σ_η^2 , σ_ε^2 . As before, define $\sigma_{\theta_t}^m = \rho\theta_t^m\sigma_\eta$. Second, there is an information transmission constraint. Each agent can observe, at most, k signals.

One way to isolate the herding effect of the model is to examine markets that are ex ante identical and show that their price levels diverge. The following proposition shows that, when there is enough information demand, information constraints force some markets to a zero-information equilibrium and create a wedge between the asset prices in markets with media coverage and markets without.

PROPOSITION 5 (Herding): *Suppose that $\tilde{m}$ markets have identical histories $(\theta^{t+1}, \varepsilon^t, x^t)$ and identical parameters and that agents can purchase at most k signals. Let $\lambda^* = \arg\max_\lambda B(\lambda)$. If $\lambda^* > k/\tilde{m}$, then the price of the risky asset is not identical across markets. The asset price is higher in the markets with higher information provision λ_{m^*} .*

When most markets are performing poorly and total information demand is low, the information transmission constraint does not bind. In these periods, the equilibrium is identical to the one-market equilibrium, where the net benefit of an additional signal purchased is zero: $B^m(\lambda^m) = 0$.

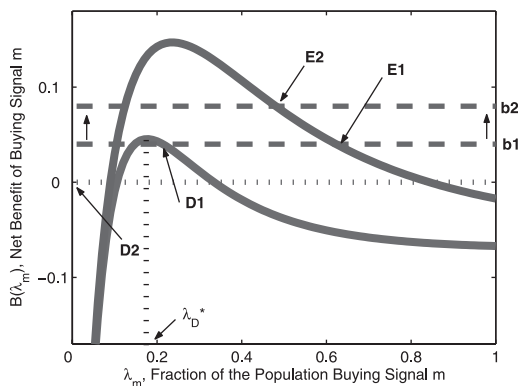


FIGURE 3. INFORMATION CROWD-OUT

Notes: Higher demand for information in a third market (not in graph) raises the net benefit cutoff level from b_1 to b_2 . Information demand in market E falls from E_1 to E_2 . Information demand in market D falls to zero.

When risk and asset payoffs are high in many markets, the information transmission constraint binds. Equilibrium information demand equates the net benefit of information purchased across all markets that supply information. Agents purchase information only in markets that can offer the equilibrium net benefit level b_t . Information trade-offs occur when the net benefit of signals in one market shifts up (either due to a reduction in the information cost or a rise in θ). This makes the information purchasing constraint bind more tightly and increases the cutoff level b_t . To maintain the higher net benefit, the information quantities in other markets must fall. A market that is near the peak of its net benefit curve λ^* may be driven to a no-information equilibrium when another market performs well, because no positive amount of information could result in a net benefit that exceeded the cutoff level b_t . Figure 3 illustrates this information crowd-out effect.

Because abundant information raises the price of an asset, information trade-offs result in asset-price trade-offs and higher cross-sectional price variance. An asset whose expected payoff θ_t and volatility σ_{θ_t} are higher than other assets generates abundant media coverage. The information raises the highest price further above the other prices. A pattern of media attention successively raising one asset price above the rest can be seen in Figure 4. In the calibrated model, the ratio of the average cross-section price vari-

ance with information to variance without information is 1.5.

V. Bringing the Model to Data

A. Data

To assess the magnitude of the model's frenzies and test the model's predictions requires data on equity markets and financial news. The data form a panel consisting of weekly observations of a price index, total return index, and the number of news stories pertaining to 23 emerging markets between 1989 and 2002 (12,217 observations). Table 2 contains descriptive statistics for the news and price data.

The price and return indices are from the S&P 500/IFCI Emerging Markets Database. The price index (P_{mt}) measures a country's market capitalization, and the total return index (R_{mt}) measures market capitalization plus cash dividends. Since the model's assets are two-period-lived, the total return index (price plus dividends) of an infinitely lived asset corresponds to the payoff of the model assets. Weekly observations are the Friday level of the index. All countries get an index of 100 at the time they enter the sample. Since the dividends extracted from these series are highly cyclical and contain zeros, dividend data are six-month, backward-looking moving averages (div_{mt}).

The information measure ($news_{mt}$) is a weekly count of the number of news stories from the *Financial Times* that contain the name of the country or the adjective form of that name, in the title or lead paragraph. The reason for using only the *Financial Times* is that a single news source avoids double-counting, its number of emerging market stories is not trending, and examination of a random sample showed this source to have largest the fraction of relevant stories. Out of a random sample of 100 *Financial Times* stories that satisfied the selection criteria, 97 contained some information related to the strength of the emerging market or the value of its assets. The reason for using emerging-market rather than firm-level data is that selecting stories related to a country's economy is easier than selecting stories for a given firm. Which stories at the industry and country level are relevant to that firm is less clear-cut.

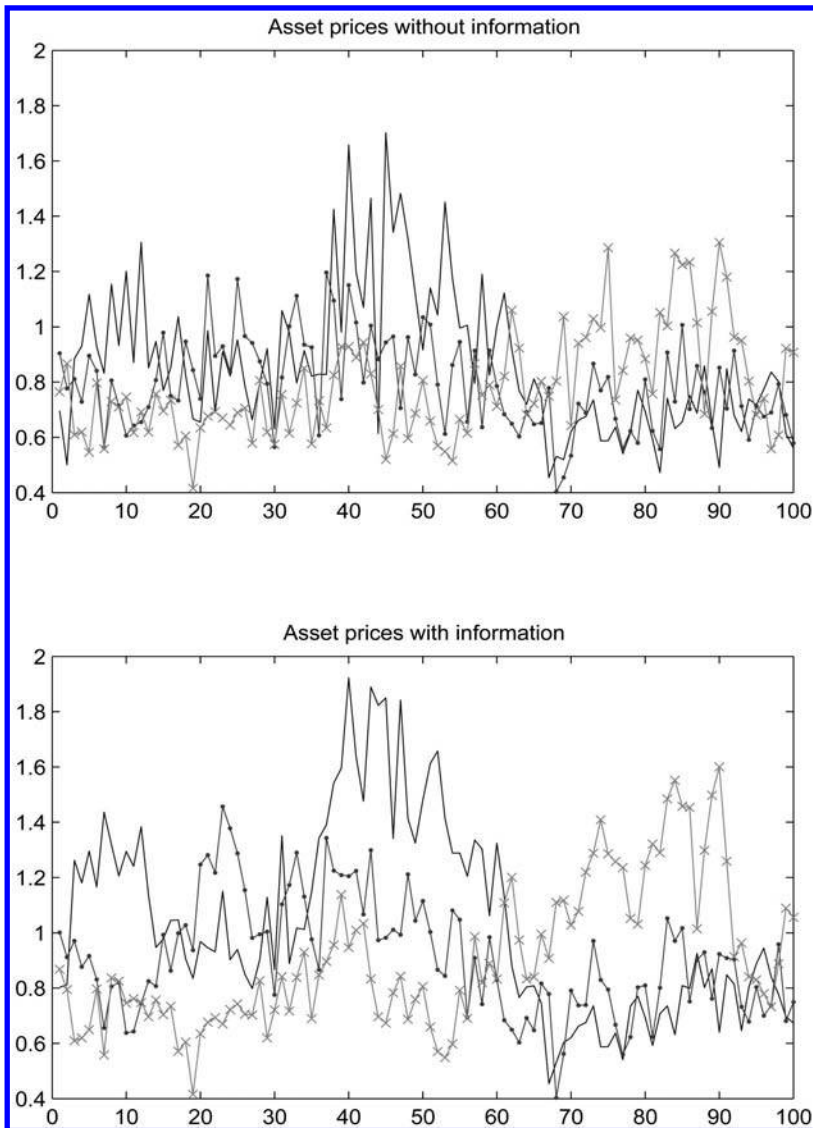


FIGURE 4. PRICE OF THE RISKY ASSET IN THREE MARKETS WHERE INFORMATION TRADE-OFFS ARE PRESENT

Notes: In the bottom panel, the information market drives a wedge between the markets covered in the media and the noncovered markets. The result is cross-sectional variance in price that is 1.5 times higher in the market with information. This simulation uses the same parameter values as the simulation in Section III. The number of signals an agent can purchase each period is one and the total number of markets is three.

B. Linking λ to News Stories Using Monopolistic Competition

A monopolistic competition model of the information market can be used to bridge the gap between the fraction of informed agents λ in the theory and the number of news stories (or sim-

ply stories) in the data. When the demand for information about a market rises, more stories are printed. This number of stories can be interpreted as varieties of the same information. Stories can also be interpreted as conveying different information, in an extension of the model.

TABLE 2—DESCRIPTIVE STATISTICS FOR DATASET

Country	Number of news stories				Asset price index		
	mean	min	max	st dev	Start date	mean	st dev
Argentina	12.5	0	71	8.5	January 1989	695.8	321.4
Brazil	19.1	0	118	10.8	January 1989	278.5	143.9
Chile	6.7	0	32	4.4	January 1989	500.0	203.8
China	50.2	14	114	14.5	September 1993	61.3	25.7
Czech Republic	9.6	1	47	5.7	April 1996	55.0	10.1
Egypt	6.4	0	47	5.7	February 1997	62.9	25.3
Greece	11.8	0	38	5.9	January 1989	378.4	216.8
Hungary	9.5	0	29	4.7	January 1994	225.1	67.0
India	25.8	2	85	9.8	November 1992	90.7	20.9
Indonesia	13.7	0	87	10.5	September 1990	67.8	39.2
Israel	22.3	6	78	9.2	August 1997	138.9	26.1
Malaysia	12.0	0	50	7.5	January 1989	174.7	82.3
Mexico	19.0	0	71	10.8	January 1989	587.6	219.4
Morocco	2.1	0	13	2.0	February 1997	135.4	24.5
Peru	4.5	0	30	3.4	September 1993	169.8	39.2
Philippines	8.6	0	44	5.8	January 1989	159.7	84.1
Poland	13.6	2	49	6.2	January 1994	601.3	145.2
Russia	51.9	17	193	20.5	February 1997	87.4	41.9
South Africa	29.3	5	71	10.9	January 1994	188.2	39.5
South Korea	21.2	0	82	11.4	January 1992	87.7	31.9
Taiwan	10.7	1	33	6.3	January 1991	126.0	31.6
Thailand	10.9	0	54	8.3	January 1989	184.3	123.7
Turkey	11.1	0	48	6.1	August 1989	208.5	116.3
All countries	15.2	0	193	13.9		255.0	237.2

Notes: Asset price data start on date listed and end in June 2002. The variable div_{mt} begins six months after the start date. News data descriptive statistics summarize the sample period from the asset price start date to June 2002.

Varieties of the Same Information.—Agent i has preferences over wealth and the type of news she receives, j :

$$(4) \quad U(W_{t+1}, j) = -e^{-a(W_{t+1} - \psi_{ij})}.$$

Each news type has the same fixed cost of production χ , and contains the same information θ_{t+1} . All types convey the same facts about an event. Buyers may have different costs of acquisition, however, or different opportunity costs for the time spent processing the information, depending on the medium or reporting style.⁵ If the consumer does not buy news, then $j = 0$ and $\psi_{ij} = 0$. Each consumer buys at most one type of news and only one unit of that type. There is a large, finite number K of news types. Consumer i 's preferences over type j are inde-

pendent across types and consumers, and are distributed uniformly: $\psi_{ij} \sim \text{Unif}(0, b)$.

Aggregate information demand is a function of the producer's information type j , as well as the set of all types produced J . Agents decide whether to produce information $d_{it} \in \{0, 1\}$, choose a type j , and charge a price c_{it} to maximize their expected profit:

$$(5) \quad \max_{d_{it}, j, c_{it}} d_{it}(c_{it}I(c_{it}, j, c_{-it}, J) - \chi).$$

Each story has the same fixed cost of production χ .

Under these assumptions, Appendix A.1 shows that the equilibrium number of stories will be proportional to $\lambda^{1/2}$:

$$q = \left(\frac{b\lambda_t}{a\chi} e^{aW_t} \right)^{1/2}.$$

⁵ Preferences over news types take the form of individual-specific costs. This assumption allows a convenient nesting of price-setting and Cournot models inside this setup. The qualitative results are the same for additive preferences over news types.

Varieties of Different Information.—Another way to relate the number of *Financial Times* stories to the model is to interpret each story as

a noisy signal about the true future state θ_{t+1} . In this setting, each agent might want multiple stories to increase the accuracy of their information. This extended model is set up in Appendix B. The intuition for the mechanism is outlined here.

Suppose that each time a supplier incurs the cost of discovery, he draws a different noisy signal of the state, a different story. Identical copies of each story can be made at zero marginal cost and sold to other agents. Another new assumption is that the cost of information discovery falls in the number of information types. If q stories are discovered, the cost of discovery is $\tilde{\chi}(q)$, $\tilde{\chi}'(q) < 0$ and $\tilde{\chi}''(q) > 0$. The declining cost captures the idea that an organization producing lots of news about a country has an infrastructure set up to collect this information, and will have an easier time marginally increasing its information production.

News stories exhibit the same price complementarities and decreasing marginal value of information as does λ , the fraction of informed investors. It is these competing forces that generate the frenzies in the model. News stories are complements because the equilibrium price of a news story falls when many are supplied. News stories are substitutes because they are correlated signals. In expectation, each observed news story provides less additional information than the previous story. When news is scarce, the decrease in the price of news is the dominant force; the net benefit of the next news story is increasing. When news is abundant, information content declines; the net benefit of additional stories falls. Because demand functions for λ and news stories have the same form, a news stories measure behaves similarly to the fraction of informed investors λ in the original model. A single demand for information λ does not exist in this model. Rather, there are identical demands λ_q for each story in positive supply. As λ_q rises, so will the equilibrium number of stories.

C. What are Financial Times Stories Capturing?

In the model, the price of information varies over time. Clearly, the price of the *Financial Times* (FT) is not variable. How does an entity like the FT fit in with a market for information? The market for information has many layers. In

the top layer are very exclusive, expensive pieces of information. In the middle are medium-cost investor newsletters. At the bottom are the types of news items that end up in the FT and on the evening news. A low-demand piece of information becomes more expensive when it stops being reported in publications like the FT and falls into the domain of expensive, specialized research. The reason the FT stops reporting obscure information is that they must provide high-demand information to remain profitable. Otherwise, subscriptions would decline and they would have to raise prices to cover their costs. Being in the FT means that a story is in high demand. Therefore, the number of FT stories is a proxy for the extent to which information about a market is easily accessible from any number of high-demand, low-cost information sources.

VI. Evidence of Media Frenzies and Herds

Three contemporaneous relationships between news and asset prices are predicted by the model: asset payoff volatility generates demand for news (Corollary 1); news increases asset price (Proposition 4); and news increases price dispersion across markets (Proposition 5). All three are supported by the data. The exogenous force driving news provision is payoff risk, which in the data is the volatility of current and future expected dividends. News and payoffs then jointly determine the price level.

A. Asset Price Volatility and News

Previous work in empirical finance (Richard Roll, 1988; Mark L. Mitchell and J. Harold Mulherin, 1994) has viewed news as an exogenous process that moves asset prices. One of the contributions of this work is to show that news should be thought of as an endogenous variable that interacts with asset markets. Corollary 1 shows that the endogeneity should take the following form: markets with higher asset payoff volatility σ_{θ_i} should generate more news, holding their parameters constant.⁶ Higher information demand comes from three sources: high payoff levels, an increase in log payoff volatil-

⁶ In fact, an increase in any parameter that would be market specific (not a preference parameter for investors that would apply equally to all markets) and would increase price volatility would also increase information demand.

TABLE 3—ASSET MARKET RISK INCREASES NEWS PROVISION

Dependent variable	Independent variable	Regression coefficient	With dividend instruments
$news_{mt}$	log payoff volatility	58.40	196.81
	$\Delta \log(R_{mt})^2$	(12.08)	(11.44)
$news_{mt}$	payoff volatility	1.93	10.73
	$(R_{mt} - R_{m(t-1)})^2 \times 10^{-4}$	(0.23)	(1.45)
$news_{mt}$	payoff	3.26	5.07
	$R_{mt} \times 10^{-3}$	(0.27)	(0.42)
mean ($news_{mt}$) in market m	log payoff standard deviation	2.45	
$(1/T) \sum_{t=1}^T news_{mt}$	$((1/T) \sum_{t=1}^T (\log(R_{mt}) - \log(\bar{R}_m))^2)^{1/2}$	(0.39)	

Notes: Regression coefficients in rows 1–3 result from estimating (6) with seemingly unrelated regressions. The coefficient estimates with dividend instruments result from 3SLS using the following instruments: div_{mt} , $(\Delta div_{mt})^2$, and four lags of the independent variable. Row 4 contains an OLS estimate of equation (7). SUR estimates use all available data, starting at the start date for each country listed in Table 2 and ending in June 2002: 12,194 observations. The instrumented regressions use div_{mt} , which begins six months after the country start date: 11,596 observations. All tables show standard errors in parentheses.

ity, and an increase in total volatility. Higher payoff levels mean that there is more value at risk in holding the asset; higher log volatility means that the asset is more volatile in the traditional sense (has bigger percentage changes in payoffs); and total volatility is the combination of these two effects. All three effects increase risk, make information more valuable, and increase information demand.

Table 3 uses panel data to estimate the contemporaneous relationship between payoff risk and information with seemingly unrelated regressions:

$$(6) \quad news_{mt} = \tilde{c}_m + \tilde{\beta}(\Delta \log(R_{mt}))^2 + \tilde{\varepsilon}_{mt}.$$

Results in rows 2 and 3 use payoff volatility and payoff, in place of log payoff volatility, as the independent variable. The coefficient estimates show that all three measures of risk covary positively with news. Row four of Table 3 isolates the cross-country effect by estimating

$$(7) \quad \frac{1}{T} \sum_{t=1}^T news_{mt} = c + \beta \left(\frac{1}{T} \sum_{t=1}^T (\log(R_{mt}) - \log(\bar{R}_m))^2 \right)^{1/2} + \varepsilon_m$$

(OLS estimation, 23 data points). The estimate shows that the mean number of news stories in a country is positively and significantly correlated with the variance of the country's asset

payoffs. How much news do financial markets generate? The coefficient on log payoff volatility predicts that a 50-percent change in payoffs ($\Delta \log(R)^2 = 0.25$) generates 15 extra stories, that week, in the *Financial Times*.

An issue that arises from this estimation is that both news and payoff risk are highly persistent. Autocorrelation of errors is a concern. Estimating the relationship between log payoff volatility and news, and including a lagged residual to correct for autocorrelation, produces a coefficient estimate of 38.4 with a standard error of 4.21.⁷

While these covariances provide support for the model, they do not speak to the causal relationship between payoff volatility and news. Establishing a causal relationship requires instruments that are exogenous with respect to news. Dividends make good instruments because they are largely predetermined outcomes of previous decisions about inputs. Dividends are also good instruments because they are good predictors of payoff volatility. In the first-stage regression of the instrument set on payoff volatility, both dividends and dividend volatility are statistically significant (P -values > 99 percent). In contrast to firm dividends, country-level dividend measures change frequently. With raw dividends, change occurs in 90 percent of observations. For the div_{mt} variable (defined in Section VI as a moving average of

⁷ The following equation is estimated by nonlinear least squares: $news_{mt} = c_m + \beta \Delta \log(R_{mt})^2 + \rho(news_{m(t-1)} - \beta \Delta \log(R_{m(t-1)})^2) + \varepsilon_{mt}$.

dividends), a change occurs in more than 99 percent of the observations. The average weekly log change in div_{mt} and payoffs R_{mt} are roughly the same size (3.8 percent and 3.7 percent, in absolute value). Therefore, payoff volatility is instrumented with div_{mt} , the volatility of div_{mt} , and four lags of payoff volatility (Table 3, column 2).⁸ The effect of asset market volatility on news becomes even larger.

One might argue that dividends do depend on news because they respond to changes in future expected cash flows, and therefore to news about future economic conditions. Changes in fundamentals that affect cash flows, however, are exactly what dividend changes are intended to capture. What is important is that printing news in the newspaper does not influence dividends. Dividends are good instruments not because they are unaffected by news content, but because they are not dependent on news provision.

The goal of this exercise was to show that asset markets generate news, in the way predicted by the theory. While the correlation results cannot speak to causality, the instrumental variables approach does suggest that asset payoff volatility causes news to be provided. Another piece of evidence that asset market volatility generates news is a Granger causality test. Since news cannot be generated by events that have not yet occurred, any relationship between past news and current prices should reflect the effect of news on price. I test the null hypothesis that lagged payoff volatility does not predict news, after controlling for lagged news. A Granger causality test with 25 lags strongly rejects this null hypothesis, with a P -value of 5×10^{-11} (F -stat 4.04). These results tell us that news is not an exogenous variable.

B. News and Asset Price Levels

When countries appear in the news more often, they should have higher asset prices, on average, because the information conveyed in the news stories decreases the investment risk. In contrast, theories of herds that rely on information cascades (Ivo Welch, 1992; Abhijit V. Banerjee, 1992; Sushil Bikhchandani et al.,

TABLE 4—NEWS AND ASSET PRICE LEVEL

Dependent variable	SUR coefficient on $news_{mt}$
Asset price	0.28 (0.03)
Price-dividend ratio	0.18 (0.02)

Notes: Equation (8) is estimated using seemingly unrelated regressions, to allow for contemporaneous correlation across countries and heteroskedasticity. Data cover from country's start date (Table 2) to June 2002: 12,194 observations.

1992; V. V. Chari and Patrick J. Kehoe, 2003) posit that frenzies arise because too little information is revealed. Flooding investors with cheap information about the markets in question should dissolve the herd because investors no longer have to rely on observing others' actions to learn about fundamentals. However, in exactly the times when price data suggest that herding may be occurring (when we see high prices and high price dispersion), the news data tell us that information is plentiful.

To distinguish between media frenzies and information cascade herds, the next exercise looks for evidence of a positive contemporaneous relationship between news and prices. To avoid the problem of more valuable assets causing more news to be provided, the relationship between news and price-dividend ratios is estimated as well. Because price indices are not comparable across countries, a country fixed effect is included in the estimation equation:

$$(8) \quad P_{mt} = c_m + \beta news_{mt} + \varepsilon_{mt}.$$

The result (in Table 4) suggests that a one-standard-deviation increase in news stories (8.2 stories) coincides with an increase in the country's asset price index that is 1 percent of the average price level (\$2.30).

Again, the issue of autocorrelated errors, or even possibly unit root errors, arises. Here, the theory provides a simple solution. When news coverage increases, prices should rise, holding payoffs constant. This implies an increase in the price-dividend ratio (P/D). Using P_{mt}/div_{mt} as the dependent variable, instead of price, produces a strong positive relationship between news and P/D (Table 4, row 2). The resulting

⁸ Results are qualitatively unchanged if lags are removed as instruments.

errors are not serially correlated (Durbin-Watson statistic = 1.88).

Is it possible that some external force is moving price and news up and down together? Unfortunately, this cannot be ruled out; there is no valid instrument for news. A valid instrument would affect news, but not asset prices. The theory is explicitly about asset payoff-relevant news. This news is, by definition, not independent of the asset price. It is unlikely, however, that newsworthy events are also events that consistently increase asset prices. If the press consistently reported information that caused asset prices to climb, price would have an upward trend. Yet price is stationary (Appendix C.3). Furthermore, if future news is expected to be good, rational investors would incorporate this expectation in their current beliefs. With more optimistic beliefs, news would not be better than expected and would not raise prices, on average. Systematic price increases caused by news content, rather than news provision, are therefore inconsistent with both the data and investor rationality.

Finally, news and price might covary because price covaries with payoff risk, which generates demand for news (Section VIA). While this may be true, three pieces of evidence suggest that there is still some causal effect of news on price. First, the positive correlation of news and P/D is unlikely to be due to reverse causality (see Table 4, row 2). High P/D assets should not generate more information demand because they don't pose more risk to investors. In fact, high P/D assets should be low-risk assets because high risk would reduce the asset's price. If asset market risk generated news (the effect documented in Table 3) but news did not affect asset markets, then most news would be provided about risky, low P/D assets; the correlation of news and P/D would be negative. Second, if correlation between news and prices arose only because asset risk generated news, then controlling for this risk would diminish the correlation. Appendix C shows that controlling for payoff volatility leaves the coefficient estimate of news on price largely unchanged. Finally, a Granger causality test reveals that past news (25 lags) predicts prices (P -value on the null hypothesis of exogenous prices is 1.2×10^{-6}). The fact that prices rise after news is supplied suggests that news is causing the price increase.

TABLE 5—THE EFFECTS OF NEWS ON PRICE AND OF PAYOFF VOLATILITY ON NEWS, JOINTLY ESTIMATED

Equation I	Coefficient on $news_{mt}$
Dependent variable	1.35
P_{mt}	(0.03)
Equation II	Coefficient on $(\Delta \log(R_{mt}))^2$
Dependent variable:	211.98
$news_{mt}$	(21.63)

Notes: Estimated using 3SLS. The equations estimated are (8) and (6). The following instruments are used for $(\Delta \log(R_{mt}))^2$: div_{mt} , $(\Delta div_{mt})^2$, and $(\Delta \log(R_{m(t-1)}))^2$. Data cover from six months after country's start date (Table 2) to June 2002: 11,596 observations for each equation.

The finding that news forecasts price increases also suggests a potentially profitable news-based trading strategy. In the model, no one can profit from buying high-news assets before their price rises; everyone learns news demand simultaneously and price adjusts immediately. In reality, if price reaction occurs more slowly, then quickly buying high-news assets may deliver excess returns.

Combining Effects: Simultaneous Equation Estimation.—If news is causing asset prices to rise, how big does the data say the effect might be? One way of answering this was to calibrate the model and measure the results of simulations. Another way to answer this question is to estimate the effect. To achieve the most efficient estimator, one should allow for cross-equation correlation in the residuals of two clearly related regressions: the effect of asset payoff volatility on news and the effect of news on price. Table 5 shows the results of jointly estimating (8) and (6), using dividend instruments for asset payoff volatility.

Allowing cross-equation error correlation increases the coefficient of news on price almost fivefold. The increase results from a negative correlation of estimation residuals between the price and news equations.⁹ What does this negative correlation mean? Residuals from equation (2) are shocks to news that are independent of asset payoff volatility (risk). A negative correlation means that when these non-risk-related

⁹ Residual correlations vary by country, ranging from -0.64 to 0.11 , with an average of -0.17 .

shocks are generating news, price is lower than what news alone would predict. In other words, events that increase news, but are not due to market risk, increase price much less than news provided in response to high risk. Theory predicts that news increases prices only when it reduces asset market risk. So, the conditions under which the correlation between news and price is high are consistent with the theory's predictions about when this mechanism should be at work.

To understand the potential magnitude of the news effect on asset price, consider the following example. Between 1989 and 1992, the average number of *Financial Times* news stories per week on Thailand was 3.8, and the average price index was 164. In 1996–1997, the average number of news stories was 19.8, and the average price index was 243. An increase of 16 news stories per week corresponds to a 22-point increase in the price index. If the correlations reported in Table 5 are due to forces in the model, then 28 percent of the increase in Thai asset prices was due to news and the remaining 72 percent to changes in fundamentals.

C. News and Asset Price Dispersion

According to the multiple markets model, news can drive a wedge between prices in similar asset markets. When news provision is low, the information transmission constraint does not bind, and markets have no effect on each other. It is only when total information demand is high that conditions for information crowd out and herds arise. Therefore, price dispersion and news should be positively correlated.

These results are the most direct evidence that the complementarity mechanism in information markets is at work. If the cost of information were fixed (as in Grossman and Stiglitz, 1980), then more information would not increase price dispersion. If it had a fixed cost, information would be a strategic substitute; investors would buy information that other investors were not purchasing. Information demand would be spread over all the assets. If the amount of information purchased about all assets rose, then prices would all rise together. Only when there is some force causing information to be concentrated in a few assets should we expect to see the distribution of asset prices become more uneven as information becomes

TABLE 6—TEST OF THE HERDING HYPOTHESIS:
PRICE DISPERSION INCREASES WHEN EMERGING
MARKET NEWS IS ABUNDANT

Dependent variable	Constant	OLS coefficient on news
Price dispersion $((1/M) \sum_{m=1}^M (P_{mt} - \bar{P}_t)^2)^{1/2}$	84.35 (7.78)	8.41 (0.49)

Notes: Coefficients result from OLS estimation of equation (9); 704 weekly observations, January 1989–June 2002.

more abundant. This force in the model is the complementarity that works through the market price of information.

To test this hypothesis, I regress the cross-sectional dispersion of prices each week on the week's average number of news stories across emerging markets:

$$(9) \quad \left(\frac{1}{M} \sum_{m=1}^M (P_{mt} - \bar{P}_t)^2 \right)^{1/2} = c + \beta \frac{1}{M} \sum_{m=1}^M news_{mt} + \varepsilon_t.$$

The results (in Table 6) support the theory's prediction. Restricting the data to the period in which all countries have price data (February 1997–June 2002, 282 observations) results in a smaller coefficient on news (2.82) which is still statistically significant at the 99-percent level.

These data measure price dispersion across markets and news largely about aggregate market risk. A natural question is whether news about idiosyncratic risk could also have significant effects on individual asset prices within a market. If we reinterpret information supply as liquidity provision, then the empirical literature on liquidity suggests that it can. One of the leading explanations for liquidity effects relies on asymmetric information. David Easley et al. (2002) show that a 10-percent change in the probability that a trade occurred when information was asymmetric can explain a 2.5-percent change in expected returns. When information is abundant in a low-cost source like the *Financial Times*, the risk of taking a position against a more informed trader falls and liquidity rises.

Hence, one way to interpret information markets is as a source of endogenous liquidity provision.

VII. Conclusion

Media frenzies are an abundance of information provided to investors about an asset or class of assets by a competitive market for information. The nonrival nature of information and the resulting price complementarity naturally create the conditions for information herds and frenzies to arise. Media frenzies raise asset prices by reducing the uncertainty about the asset's payoff. In a single market, media frenzies increase prices and price volatility in what appears to be an asset-buying frenzy. In multiple markets with trade-offs in information demand, a media frenzy in one market raises the market price and increases the cross-sectional price variance. The resulting price path looks like herds of investors stampeding from one market to the next.

Using information markets to generate asset market frenzies avoids some of the problems of earlier models. It avoids the criticism levied against the herding models that their discrete choice space is needed (In Ho Lee, 1993; Xavier Vives, 1993) and that market clearing prices interfere with the effect (Christopher Avery and Peter Zemsky, 1998). This bubble-like outcome does not rely on agents who are not subject to transversality conditions (as in Oliver Blanchard and Mark W. Watson, 1982), or have short trading horizons (as in Kenneth A. Froot et al., 1992). The reason that these models must make restrictive assumptions about asset markets is that they are relying on the asset market to generate their complementarity. Even Froot et al., who do generate complementarity in information acquisition, get that result because traders all want to hold the same assets. Likewise, Gadi Barlevy and Pietro Veronesi (2000) generate information complementarity when asset payoffs are low by altering the distribution of payoff shocks. In this model, the source of the complementarity is purely in the information market, where the nature of information makes complementarity a natural outcome. The asset market effects reflect only the reduced investment risk that accompanies close media scrutiny.

Complementarity of information acquisition is not limited to settings where information is

purchased in a competitive market. The idea can be extended to more informal settings as well. Consider the following example of an individual learning in the presence of network externalities. Suppose that all information is freely available. An investor must decide, however, whether to exert effort to find news that reveals asset-payoff-relevant information. Other people the investor meets can alert him to events and news of interest, reducing the required search effort to discover the related information. Such a setting could produce an information complementarity where other people knowing information makes acquiring that same information more attractive.

Complementarity of information is also not undermined when variation in signal quality is added. Suppose some investor demanded a signal of higher quality than what was currently being produced. The investor must pay, at least, the entire marginal cost of increasing the quality of the signal produced. If a second investor also wants to increase his signal quality, the higher-quality signal can now be replicated cheaply; the cost of the quality increase, per investor, will fall. Complementarity still exists because the cost of acquiring a higher-quality signal falls when others demand higher-quality information as well. It is true that all of the force of the information markets mechanism is working along the quantity dimension—how many copies of a signal are produced. But this story is exactly about why investors appear to copy the strategies of other investors. The reason we see investors copying each other's strategies is that copying their information is cheap, no matter what its quality.

Introducing cross-market information interactions into the theory could generate contagion—high correlation in countries' prices when prices are low and information is scarce. Complementarity magnifies price volatility as markets reinforce each other's booms and crashes. Information substitutability, arising from correlation in market fundamentals, generates price correlation that is strongest in crises. If market x 's signal is informative about market y , then beliefs about y react to news about x . The reaction will be strongest when information is scarce. Information is scarce when expected payoffs and price are low. Countries where the market for information is thin may be particularly susceptible to contagion

because high-variance prior beliefs allow small pieces of new information to move beliefs, and therefore prices, substantially.

Media markets could also provide insight into the home bias puzzle. In a coordination model with potential multiple equilibria, a natural question is whether focal points exist. Local companies may provide a natural focal point for news providers, prone to herding. An abundance of low-cost information available to local residents will increase their demand for the asset.

Modelling media coverage also provides a stronger microeconomic foundation for models that use time-varying information flows to generate asymmetry (Veldkamp, 2002; Stijn Van Nieuwerburgh and Veldkamp, forthcoming). Information, such as news stories, that becomes abundant when the market is strong generates sudden crashes and gradual booms. When fundamentals are strong and asset prices are high, media frenzies arise and agents become well-informed. This model provides a market mechanism that ties information flow to asset prices in the way needed to generate this asymmetry.

Media frenzies rely crucially on the high fixed cost and low marginal cost of information provision. As innovations such as the Internet reduce information's marginal cost, the theory predicts that media-driven frenzies will become increasingly common. This prediction seems in line with recent history and offers a warning about the future volatility of asset markets.

APPENDIX A

A.1. PROOF OF PROPOSITION 1:

When information demand is positive, equilibrium demand is a λ such that the net benefit of information purchase is zero. Let payoff variance be denoted $\sigma_{\theta r} \equiv \rho \theta_i \sigma_\eta$. Manipulating the net benefit equation (3) and setting the benefit of information (left side of equation (10)) equal to its cost in utility units (right side of (10)) yields the equilibrium information demand:

$$(10) \quad \frac{\sigma_x^2 (a\sigma_\varepsilon/\lambda_i)^2}{1 + (\sigma_x^2/\sigma_{\theta r}^2)(a\sigma_\varepsilon^2/\lambda_i)^2} = e^{2a(c(\lambda_i) + \tilde{y})} - 1.$$

For a derivation of this expression, see Appendix B of Grossman and Stiglitz (1980). This equilibrium condition ignores integer constraints on the number of agents.

Model I: Equilibrium Price Equals Average Cost ($c = \chi/\lambda$)

Proof: Suppose the equilibrium information price was above average cost. Then, an alternate supplier could enter the market with a slightly lower price and make a profit. If a supplier set price below marginal cost, they would make a loss. This strategy would be dominated by no information provision. If there are two or more suppliers, then either price is above marginal cost, which can't be an equilibrium by the first argument, or both firms price at (or below) marginal cost, split the market, and make a loss, which is dominated by exit.

Model II: Cournot Model

Profits from producing information depend on the quantity of information production chosen, as well as the quantities chosen by all other producers. Let the price for information, when agent i produces $\tilde{d}_{it}$ copies of news, per capita, and all other agents produce a per capita quantity $\tilde{d}_{-it}$, be $\tilde{c}(\cdot, \cdot)$. The objective of the information producer is to maximize profit:

$$(11) \quad \max_{\tilde{d}_{it}} \tilde{d}_{it} \tilde{c}(\tilde{d}_{it}, \tilde{d}_{-it}) - \chi \mathbb{1}_{\tilde{d}_{it} > 0}.$$

Given a quantity of information supplied to the market, the asset price and demand will be identical to the Bertrand case. What changes in a Cournot model is the quantity of information provided. To solve for that quantity, begin by showing that one firm always wants to maintain a monopoly position in the information market, rather than accommodate an entrant. Suppose a firm produced a quantity of information Q_1 , measured in signals per capita, such that a second producer could produce a quantity Q_2 for a profit $\pi(Q_2) \geq 0$. Then, the incumbent firm could do strictly better by producing an amount $Q_1 + Q_2$ and capturing the additional profit $\pi(Q_2) + \chi > 0$. The χ term enters because the firm does not have to bear the production fixed cost twice.

Now, the problem can be formulated as a constrained maximization where the incumbent maximizes profit, subject to the constraint that he must deter entry. Let P denote the price of information, Q_1 the incumbent's quantity of signals per capita, and Q_2 the challenger's:

$$\begin{aligned} & \max_{Q_1} P(Q_1) * Q_1 \\ & \text{s.t. } \max_{Q_2} P(Q_1 + Q_2) * Q_2 \leq \chi. \end{aligned}$$

Rearranging equation (10) yields a price-quantity relationship for information demand:

$$(12) \quad P(Q) = -\frac{1}{2a} \ln \left(\frac{Q^2 + M}{Q^2 + M + MN} \right),$$

where $M = (\sigma_x/\sigma_\theta)^2(a\sigma_\varepsilon^2)^2$ and $N = (\sigma_\theta/\sigma_\varepsilon)^2$.

The negative relationship between price and quantity comes from the demand equation (12). The negative first derivative of the demand equation is sufficient to prove the proposition.

Model III: Monopolistic Competition

Following Jeffrey M. Perloff and Steven C. Salop (1985), free-entry in the information market implies that equilibrium information price $c(\lambda_i)$, times the number of agents who buy information $N\lambda_i$, times the probability an agent most prefers information of type j , H_j , must be equal to the fixed cost of entry:

$$c(\lambda_i)\lambda_i H_j = \chi.$$

As in Perloff and Salop, I ignore the integer constraints on the number of news types. In equilibrium, each type j attracts an equal share of the market. Let n_j be the number of types supplied in equilibrium:

$$c(\lambda_i)\lambda_i = \chi n_j.$$

The first-order condition of the news supplier's profit maximization problem yields $p = b/n_j$. Combining this with the free-entry conditions, we get equilibrium price and number of firms:

$$\begin{aligned} n_j &= \left[\frac{b\lambda_i}{\chi} \right]^{1/2} \\ p &= \left[\frac{b\chi}{\lambda_i} \right]^{1/2}. \end{aligned}$$

The equilibrium price is decreasing in λ_i . Thus, the more people buy news, the lower the price of news will be.

A.2. PROOF OF PROPOSITION 2: Information Complementarity

$B(\lambda; \theta_i)$ is the net benefit of buying a signal. Grossman and Stiglitz (1980, Appendix B) show that equation (1) can be expressed as

(13)

$$\begin{aligned} B(\lambda; \theta_i) &= \left[1 + \frac{\sigma_x^2(a\sigma_\varepsilon/\lambda_i)^2}{1 + (\sigma_x^2/\sigma_\theta^2)(a\sigma_\varepsilon^2/\lambda_i)^2} \right]^{1/2} \\ &\quad - e^{2a(c(\lambda_i) + \tilde{\psi})}. \end{aligned}$$

To begin, assume that $\tilde{\psi} = 0$ (as in Bertrand or Cournot markets). Differentiation with respect to λ yields

$$\begin{aligned} & \frac{\partial B(\lambda; \theta_i)}{\partial \lambda} \\ &= -\lambda a^2 \sigma_x^2 \sigma_\varepsilon^2 \frac{[\lambda^2 + a^2 \sigma_x^2 \sigma_\varepsilon^2 (1 + \sigma_\varepsilon^2/\sigma_\theta^2)]^{-1/2}}{[\lambda^2 + (\sigma_x^2/\sigma_\theta^2)(a\sigma_\varepsilon^2)^2]^{3/2}} \\ &\quad - ac'(\lambda)e^{ac(\lambda)}. \end{aligned}$$

The first term is zero when $\lambda = 0$ and is negative for all $\lambda > 0$. This is because the benefit of information declines as the price level becomes more informative. According to Proposition 1, $c'(\lambda) < 0$. Thus, the second term is strictly positive for all $\lambda \geq 0$. Since $\partial B(\lambda; \theta_i)/\partial \lambda$ is continuous in $\lambda > 0$, and is strictly positive in the neighborhood of zero, the proposition follows.

When $\tilde{\psi} \neq 0$, there is a second positive externality of an increase in demand for news: increased diversity of available news types. Note that n_j in the proof of Proposition 1 is increasing in λ . The expected lowest individual-specific cost of information for a given agent i is

$$E[\tilde{\psi}] = \frac{b}{n_j + 1}.$$

As the number of news types n_j increases, $E[\tilde{\psi}]$ decreases. Every agent must be made weakly better off with more available news types to choose from. Therefore, the second term in the derivative of the net benefit equation with news types is

$$-a(c'(\lambda) + \partial \tilde{\psi}/\partial \lambda)e^{2a(c(\lambda_i) + \tilde{\psi})}$$

which is still strictly increasing near zero.

A.3. PROOF OF PROPOSITION 3: Information Is Provided when Asset Values Are High

Step 1: If $B(\lambda) > 0$ holds for any $\lambda > 0$, then there exists an equilibrium $\lambda^e > 0$. A level of information demand, λ^{es} , is an equilibrium if the people who buy information weakly prefer to do so ($B(\lambda^e) \geq 0$), and either $\lambda^e = 1$, or an uninformed person weakly prefers staying uninformed ($B(\lambda^e + \varepsilon) \leq 0$, $\varepsilon > 0$).

Suppose that there is some λ such that $B(\lambda) > 0$ but no equilibrium exists. If λ is not an equilibrium, then it must be that $B(\lambda + \varepsilon) > 0$. By induction, if $B(\lambda + \tilde{\varepsilon})$ is not an equilibrium, for any $\tilde{\varepsilon} > 0$, it must be that $B(1) > 0$. But then $\lambda = 1$ is an equilibrium, which is a contradiction.

Step 2: A positive information equilibrium exists if and only if

(14)

$$\left\{ \sigma_{\theta_{t+1}}^2 + \sigma_{\varepsilon}^2 + \sigma_{\theta_{t+1}}^2 \left[1 + \left(\frac{a\sigma_{\varepsilon}\sigma_x}{\lambda_t\sigma_{\theta_{t+1}}} \right)^2 \right]^{-1} \right\}^{1/2} \\ \geq \exp[a(c(\lambda_t) + \tilde{\psi})]\sigma_{\varepsilon} \quad \text{for some } \lambda_t \in (0, 1).$$

Using step 1 and the Grossman-Stiglitz result (p. 405) that $\text{Var}[u|P_{\lambda}] = \sigma_{\theta}^2 + \sigma_{\varepsilon}^2 - \{\sigma_{\theta}^2/[1 + (a\sigma_{\varepsilon}^2/(\lambda\sigma_{\theta}))^2\sigma_x^2]\}$, inequality (14) follows.

Step 3: There exists a θ^* such that for all $\theta \geq \theta^*$, (14) holds. Only two terms in (14) vary: λ and σ_{θ} . Let θ^* be the lowest $\theta > 0$ such that (14) holds for some λ . Let the λ that satisfies (14) at θ^* be λ^e .

Partial differentiation shows that the left side of (14) is increasing in σ_{θ} . Furthermore, σ_{θ} is increasing in θ , when $\theta > 0$. Thus, for any $\theta \geq \theta^*$, (14) holds at $\lambda = \lambda^e > 0$.

Step 4: For $\theta_t > \theta^*$, an equilibrium where $\lambda_t = 0$ does not exist. Suppose $\lambda_t = 0$ was an equilibrium. Then, some agent could decide to discover information at a per capita cost χ and sell it at a price $\chi/\lambda^e + \varepsilon$. Since $\theta_t > \theta^*$, the net benefit curve lies strictly above zero at $\lambda^e - \varepsilon$ and the information producer can capture positive surplus by raising price. Therefore, zero information production cannot be an equilibrium strategy for all agents.

Step 5: When $\theta_t \leq \theta^*$, the equilibrium is $\lambda_t = 0$. If (14) does not hold for any $\lambda > 0$, then there cannot be an equilibrium where $\lambda^e > 0$. If there were, then all agents who purchase information would be strictly better off not purchasing it.

A no-information equilibrium exists because neither purchasers nor suppliers want to deviate from their equilibrium strategies. No purchaser would purchase information; (14) rules this out. No supplier would want to supply information because at any price greater than average cost, the demand for the information would be zero.

A.4. PROOF OF COROLLARY 1: Payoff Variance Increases Information

Recall that σ_{θ_t} is linear in θ_t , so that $\forall \theta_t > 0$, σ_{θ_t} is increasing in θ_t . From Proposition 4, we know that for $\theta_t < \theta^*$, $\lambda_t = 0$. For $\theta_t \geq \theta^*$, $\lambda_t > 0$. In a positive information equilibrium, information demand is determined implicitly by equation (10). Applying the implicit function theorem to $B(\lambda) = 0$ (equation (13)) in a region where $B'(\lambda) \geq 0$, which is a necessary condition for a stable positive-information equilibrium, yields $\partial\lambda_t/\partial\sigma_{\theta_t} > 0$.

A.5. PROOF OF PROPOSITION 4: Information Increases Asset Price

Begin by taking the expectations of asset demand, conditional on θ_t . For agents who purchase information, expected demand is

$$\frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\sigma_{\varepsilon}}.$$

For agents who do not purchase information in the positive information equilibrium,

$$\frac{E[E[\theta_{t+1}|P_t, \theta_t]|\theta_t] - rE[P|\theta_t]}{a\text{Var}[u_{t+1}|P_t]} \\ = \frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\text{Var}[u_{t+1}|P_t]}$$

by iterated expectations. Thus, expected total demand in the information equilibrium is

$$\lambda \frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\sigma_{\varepsilon}^2} \\ + (1 - \lambda) \frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\text{Var}[u_{t+1}|P_t]}.$$

Note that the variance term $\text{Var}[u_{t+1}|P_t]$ is not stochastic, conditional on θ_t , because it depends only on parameters that can be deduced from θ_t .

In the no-information equilibrium, let P^U denote the “uninformed” asset price. Then, expected demand is:

$$\frac{E[\theta_{t+1}|\theta_t] - rE[P^U|\theta_t]}{a\sigma_\varepsilon^2 + \sigma_\theta^2}.$$

Since demand must be equal to x_t in both equilibria, the two expressions for demand must equal each other:

$$\begin{aligned} \lambda \frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\sigma_\varepsilon^2} \\ + (1 - \lambda) \frac{E[\theta_{t+1}|\theta_t] - rE[P|\theta_t]}{a\text{Var}[u_{t+1}|P_t]} \\ = \frac{E[\theta_{t+1}|\theta_t] - rE[P^U|\theta_t]}{a\sigma_\varepsilon^2 + \sigma_\theta^2}. \end{aligned}$$

The fact that information decreases the variance of the asset payoff implies that $\sigma_\varepsilon^2 + \sigma_\theta^2 > \text{Var}[u_{t+1}|P_t] > \sigma_\varepsilon^2$. This implies that $E[P|\theta_t] > E[P^U|\theta_t]$.

The fact that $\text{Corr}(P_t, u_{t+1}|\theta_t)$ is increasing in λ means that $\text{Var}[u_{t+1}|P_t]$ is decreasing in λ . Following the same steps as above, one can show that if $\lambda_1 > \lambda_2$, then $E[P_1|\theta_t] > E[P_2|\theta_t]$.

A.6. PROOF OF COROLLARY 2: Frenzies

The result follows directly from Propositions 3 and 4.

A.7. PROOF OF PROPOSITION 5: Herding

Suppose that $\tilde{m}$ markets have identical histories $(\theta^{t+1}, \varepsilon^t, x^t)$ and identical parameters, and that agents can purchase at most k signals. Let $\lambda^* = \text{argmax}_\lambda B(\lambda)$. If $\lambda^* > k/\tilde{m}$, then the price of the risky asset will not be identical across markets. It will be higher in the market with higher information demand λ_{mr} .

Suppose not. Suppose that the price of the risky asset is identical in all $\tilde{m}$ markets. From Proposition 4 we know that information increases the asset price. So for each market to have the same asset price, given the same histories, they must have the same amount of information as well. Suppose $\lambda = 0$ in all markets. For this to be an equilibrium, it must be that $\text{argmax}_\lambda B(\lambda) = 0$. But then $\lambda^* = 0 < k/\tilde{m}$,

which contradicts one of the conditions of the proposition. Therefore, all markets have the same positive amount of information λ_m . The information capacity constraint says that the maximum fraction of informed agents in each market would be $\lambda_m = k/\tilde{m} < \lambda^*$.

Since each market has a $\lambda_m < \lambda^*$, the net benefit of information in each market is increasing in λ_m . This means an agent in one market, who gets a net benefit of $B(\lambda_m)$ from his information, has an incentive to purchase a signal in another market to get a net benefit of $B(\lambda_m + \varepsilon) > B(\lambda_m)$. Therefore, this cannot be an equilibrium. The fact that the price will be higher in the market with the higher λ_m follows directly from Proposition 4.

APPENDIX B: AN EXTENDED MODEL WITH NEWS STORIES

This section describes a model of news supply where news stories are varieties of different information. The equilibrium number of stories n_j increases with the fraction of informed agents λ in the original model.

Investors, preferences, asset markets, and asset payoffs are as described in Section I. Only information markets differ. For a cost $\tilde{\chi}(n_j)$, each information supplier j discovers a different story:

$$q_{jt} = \theta_{t+1} + \gamma_{jt} \quad \gamma_{jt} \sim N(0, \sigma_\gamma^2)$$

and sells stories—identical copies of q_{jt} —at an endogenous price: $\tilde{\chi}'(n_j) > 0$. Any agent can become a supplier at any time. Suppliers are price-setting monopolistic competitors. Agents can buy multiple stories.

Consider an agent who has information $\mathcal{F}_{it}$, either purchased or inferred from the price level that leads her to believe the following: $E[u_{t+1}|\mathcal{F}_{it}] = \hat{\theta}$ and $\text{Var}[u_{t+1}|\mathcal{F}_{it}] = \tilde{\sigma}_j$. Suppose the agent buys an additional story j . When no other agents buy j ($\lambda_j = 0$), the price level is uninformative about q_j . Define $\tilde{\sigma}_{\theta j}$ to be the reduction in payoff variance after observing q when $\lambda_q = 0$. Define $\tilde{\sigma}_{\varepsilon j}$ to be the payoff variance conditional on $\mathcal{F}_{it}$ and q_j . $\tilde{\sigma}_{\varepsilon j}^2 = \text{Var}[u_{t+1}|\mathcal{F}_{it}, n] = \tilde{\sigma}_j^2 - \tilde{\sigma}_{\theta j}^2$. Holding these conditional variances fixed, the decision of whether to buy story j takes the same form as that in the original model. The net benefit of buying j when $N\lambda_j$ investors also buy j is

$$(15) \quad \tilde{B}(\lambda_j, n_j) = \frac{\sigma_x^2(a\tilde{\sigma}_{ej}/\lambda_{jt})^2}{1 + (\sigma_x^2/\tilde{\sigma}_{\theta j}^2)(a\tilde{\sigma}_{ej}/\lambda_{jt})^2} - e^{2a\tilde{\chi}(n_j)/\lambda_{jt}} + 1.$$

What changes when multiple noisy signals—stories—are introduced is that $\tilde{\sigma}_{\theta j}$ changes for each story. Whenever a new story is supplied to the market, it is partially revealed through the equilibrium price level. Both agents who purchase the story, and those who don't, have a lower variance estimate of the payoff.

The net benefit of purchasing the next story increases, due to price complementarity, and then decreases as the price becomes more informative. The parallel between the net benefit for the number of stories and λ in the original model causes their equilibrium quantities to move together.

The force that causes the net benefit of the next story to decrease is more substitutable information. As the number of stories grows, each new story reduces payoff variance less: $\tilde{\sigma}_{\theta j}$ falls. The reason is that in the Bayesian updating formula, a new signal (the story) with the same variance is given less weight when the variance of the prior is less. For example, if the variance of a normally distributed prior is $\tilde{\sigma}_j$ and the variance of the signal is σ_γ , then the variance of the posterior is $\tilde{\sigma}_j\sigma_\gamma/(\tilde{\sigma}_j + \sigma_\gamma)$. The variance reduction from observing the story is $\tilde{\sigma}_{\theta j} = \tilde{\sigma}_j(1 - \sigma_\gamma/(\tilde{\sigma}_j + \sigma_\gamma))$, which is increasing in σ_γ . So, the larger the prior variance σ_j , the larger the variance reduction $\tilde{\sigma}_{\theta j}$. Because each new story reduces the variance of beliefs that will be the prior beliefs on observing the next story, each new story reduces $\tilde{\sigma}_{\theta j}$.

The force that causes the net benefit of stories to rise is price complementarity. Complementarity in stories arises because each story lowers the fixed cost of discovering all stories. Since the average cost of a story falls, the equilibrium price of stories falls too. This follows from the proof of a declining price in a monopolistic competition model (Appendix A.1). Given that n_j stories will be produced, consumers view these stories as ex ante identical. In equilibrium, demand for each story produced λ_j must be the same for all stories. Appendix A.1 shows that

the number of varieties offered by competitive monopolists increases in the demand for each variety. This tells us that n_j increases in λ_j . To the extent that λ_j in this setting is the equivalent of λ in the model with one news story, the number of stories is increasing with λ .

APPENDIX C: DATA

C.1. News and Asset Data: Descriptive Statistics

News counts are the number of stories in the *Financial Times* in a given week, with the country name in the title or lead paragraph of the article. For every country, this series begins in 1/1989 and continues until 6/2002 (703 observations per country). Asset price index is the S&P 500/IFCI investible index from their Emerging Markets Database. Price and payoff (total return) data are an unbalanced panel. Many of the markets were not opened to foreign investment until after 1989. The date at which price and payoff indices become available for a country is listed as the start date in Table 2. Price and payoff indices are normalized to 100 at the country start date.

C.2. Robustness Checks

Proposition 4 shows that news increases price by reducing the conditional risk of investment. The model also predicts, however, that the price level can affect news if it is correlated with price volatility, and therefore payoff volatility in the data. To check if the correlation between price and news is robust to price and payoff volatility, I reestimate (8) including price volatility as a control variable. (Seemingly unrelated regression: 12,194 observations, country start date – 6/2002.)

$$(16) \quad P_{mt} = \alpha_m + \beta_1 news_{mt} + \beta_2 (\log(R_{mt}) - \log(R_{m(t-1)}))^2 + \varepsilon_{mt}.$$

This inclusion has a negligible effect on the parameter estimates. The coefficient on price, which was 0.28 before (Table 4), rises to $\beta_1 = 0.30$ (White standard error is 0.04). The volatility coefficient is $\beta_2 = -190.9$ (33.8).

Similarly, I reestimate the simultaneous equations system controlling for news variance

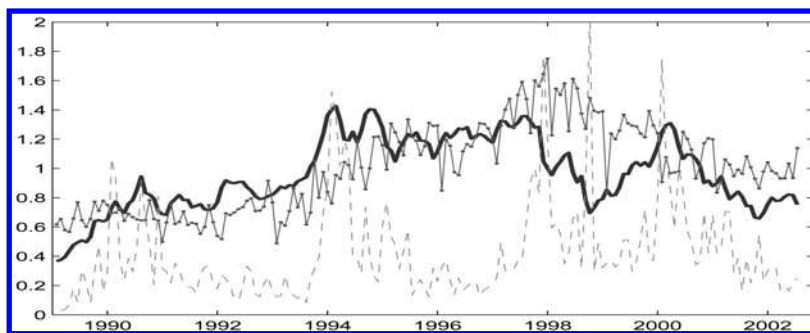


FIGURE 5. AVERAGE PRICE INDEX (SOLID LINE), NUMBER OF NEWS STORIES (DOTS), AND PAYOFF VOLATILITY (DASHED LINE)

Notes: Monthly data. All series are normalized by their respective country means before they are averaged across all emerging markets.

in the news equation and payoff variance in the price equation: equation (16) and

$$(17) \quad \text{news}_{mt} = \alpha_m + \beta_3 (\Delta \log(R_{mt}))^2 + \beta_4 (\Delta \log(\text{news}_{mt}))^2 + \varepsilon_{mt}.$$

The coefficient of news on price β_1 was 1.35 in Table 5, and rises to 2.04 (0.02) after controlling for payoff volatility. The coefficient of payoff volatility on news β_3 was 211.98 in Table 5; the same coefficient is 58.10 (4.37) after controlling for news volatility. Although one effect becomes stronger and the other weaker, neither changes sign or becomes insignificant.

C.3. Unit Root Tests

Stationarity of the data is a cause for concern. Price data are usually trending and news stories may well be proliferating. This section shows that these concerns are not warranted in this dataset.

The news data are the number of stories published per week in one news source: the *Financial Times*. Figure 5 plots the average number of news stories per emerging market. The number of stories in 2002 is approximately the same as in 1994, but is higher than in the early 1990s. An augmented Dickey-Fuller test of each country's news series rejects the null hypothesis of a unit root, at the 1-percent significance level, for every country but one—Korea. A unit root for Korean news can be rejected at the 5-percent level.

The price data are an emerging market equity price index. The units for all countries indices are U.S. dollars. The solid line in Figure 5, representing the average price index level in all emerging markets, has no obvious trend. Augmented Dickey-Fuller tests of individual countries' price series reject the null hypothesis of a unit root at the 5-percent level for one country, and at the 10-percent level for five countries. However, a more powerful test would use all countries' data in a joint stationarity test.

One way to jointly test countries' price data for stationarity is to estimate

$$\Delta P_{mt} = \sum_{i=1}^q \phi_i \Delta P_{m(t-i)} + \alpha + \rho P_{m(t-1)} + \varepsilon_{mt}$$

and test the null hypothesis $H_0 : \rho = 0, \alpha = 0$. OLS estimation with a t-test rejects the null hypothesis at the 1-percent level. It rejects the simple null $\rho = 0$ at the 1-percent level as well. Estimating by SUR to allow for correlated country shocks and heteroskedasticity yields the same results. The same system is estimated, allowing ϕ and α to be country-specific, as in Andrew Levin et al. (2002). A likelihood ratio test rejects both $H_0 : \rho = 0, \alpha = 0$ and $\tilde{H}_0 : \rho = 0$ at the 1-percent level.

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